Survival Outcomes of AIDS Patients in Australia

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Outline

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Introduction and Objective

AIDS

Acquired Immunodeficiency Syndrome (AIDS), a disease that affects the immune system, reduces the body's ability to fight infections. Studying survival after diagnosis helps us understand how long patients live and which factors affect their risk of death.

Main Objective

- To investigate the survival outcomes of AIDS patients in Australia.
- To understand how long patients survive after diagnosis.
- To identify factors that influence the risk of death.

Methodology

Methods

This study used survival analysis methods to examine survival outcomes of AIDS patients in Australia.

- Kaplan–Meier survival estimation
- Log-rank test
- Cox proportional hazards model
- Proportional hazards assumption test

Dataset and Survival Variables

Dataset: `Aids2`

- Number of patients: **2843**
- Main variables: age, sex, state, transmission category, diagnosis date, death date, and status

Survival variables

$$\text{Survival time} = \text{Death date} - \text{Diagnosis date}$$

$$\text{Survival years} = \frac{\text{Survival time}}{365.25}$$

- Event = 1 if death occurred
- Event = 0 if censored

Age Distribution of AIDS Patients

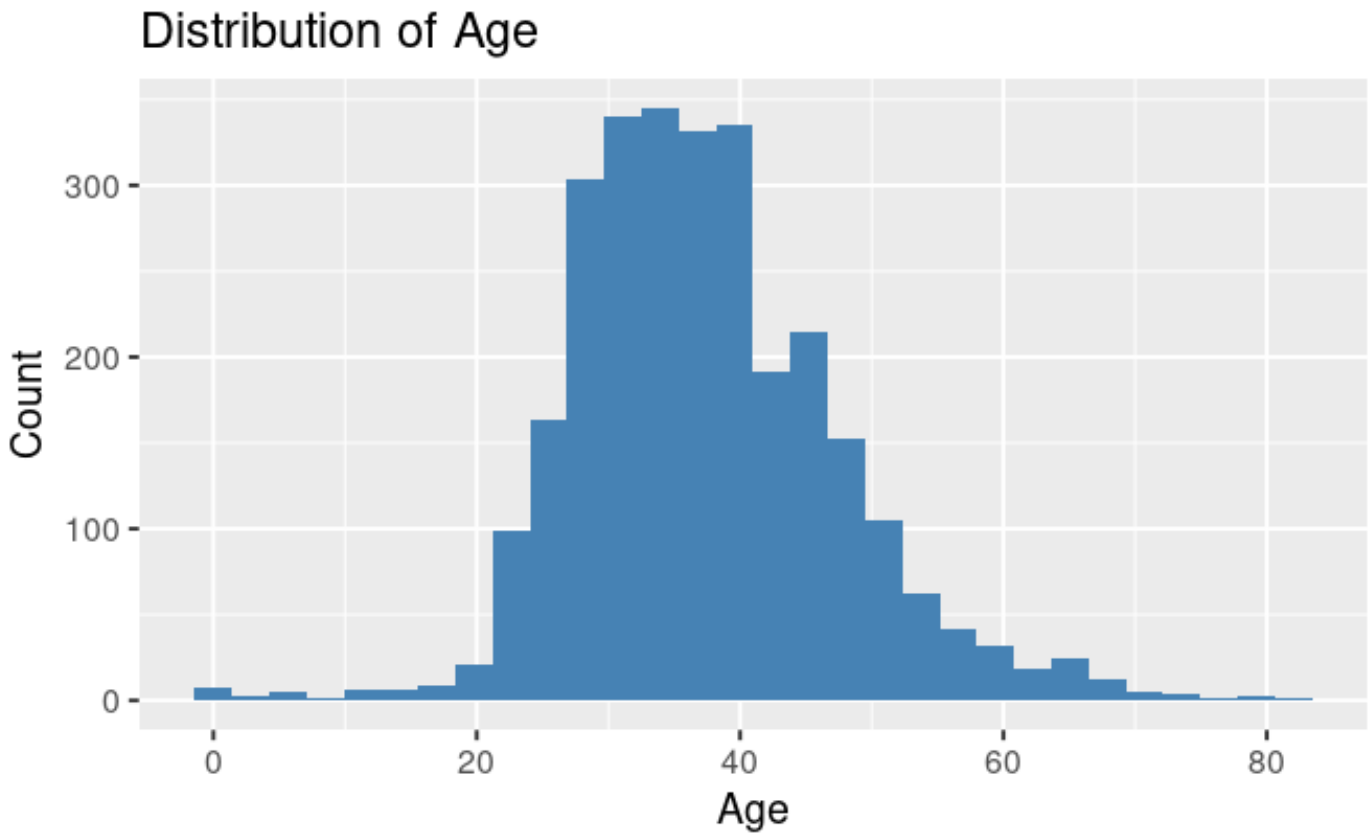

Figure 1: Age distribution of AIDS patients in Australia

- The histogram shows the distribution of age among AIDS patients.
- Most patients are between 25 and 45 years old.
- This shows that most patients are young and middle-aged.

Sex Distribution of AIDS Patients

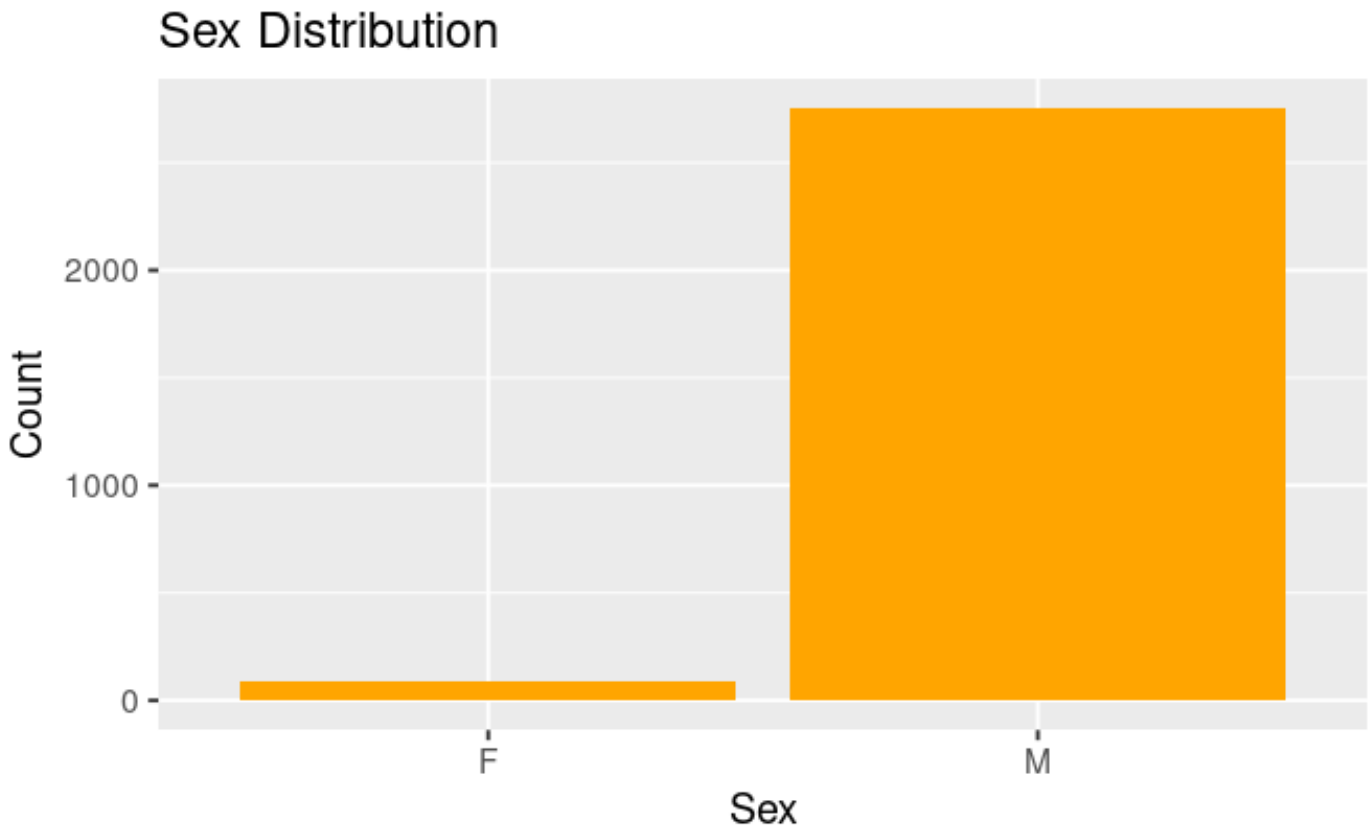

Figure 2: Sex Distribution of AIDS patients in Australia

- The number of male patients is much higher than female patients.
- Only a small number of females are in the dataset.
- This means most AIDS patients in this study are male.

Kaplan–Meier Survival Curve

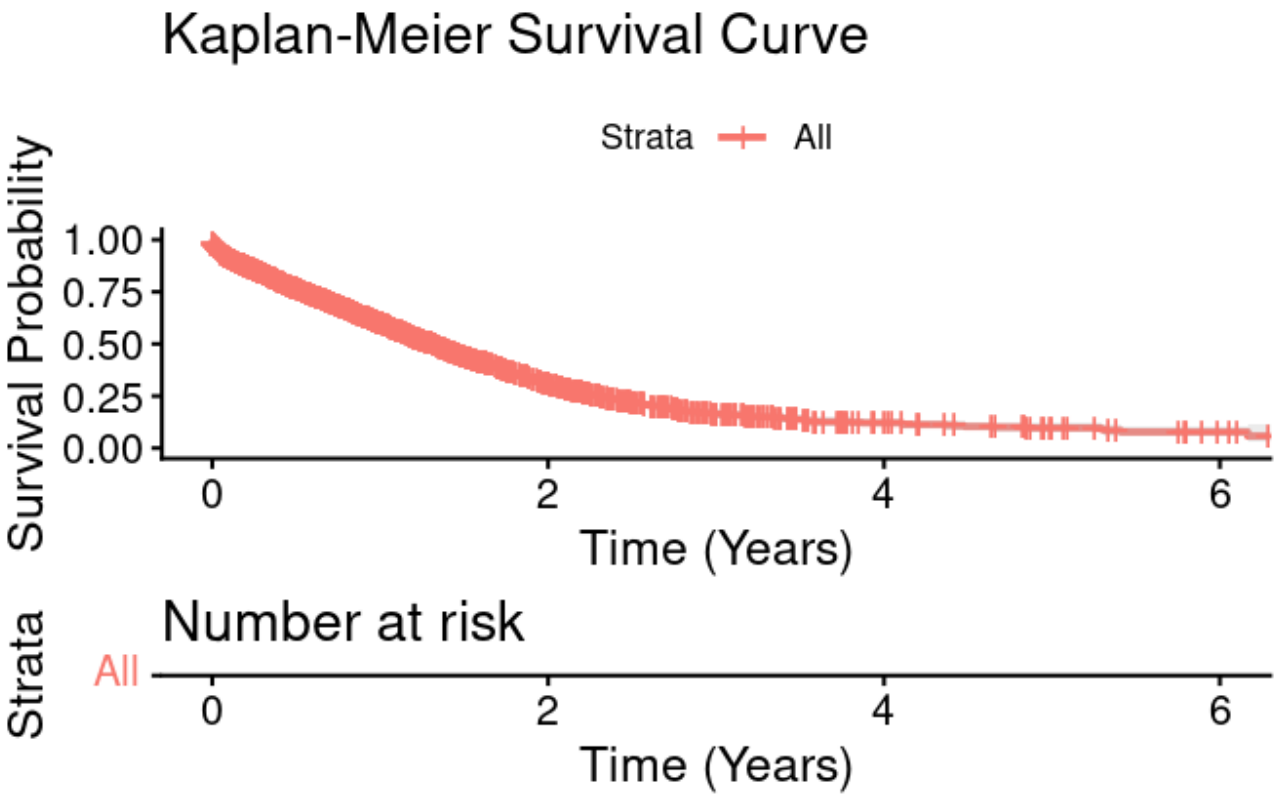

Figure 3: Kaplan–Meier survival curve for all AIDS patients

As time passes, the curve goes down, which means more patients die during follow-up. After 4 to 6 years, only a small number of patients remain alive.

Survival Curve by Sex

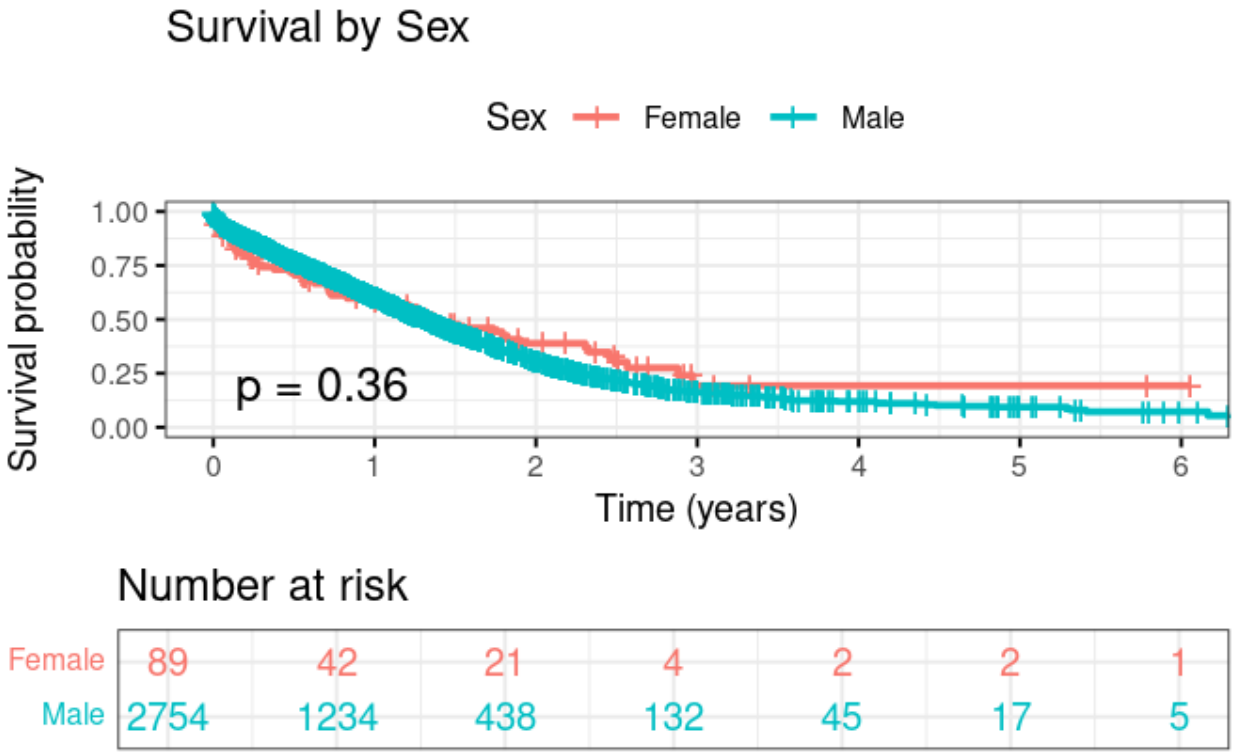

Figure 4: Kaplan–Meier survival curves by sex

- The p-value of 0.36 shows that survival is similar for males and females.
- The female curve is a little higher than the male curve, females may survive a little longer.

Cox Proportional Hazards Model

Table 1: Cox Proportional Hazards Model

Variable	Hazard Ratio	p-value
Age	1.013	< 0.001
Sex (Male)	0.954	0.790
Transmission: id	0.610	0.045
Transmission: het	0.477	0.005
Transmission: blood	1.42	0.010

Interpretation:

- Age strongly affects the risk of death.
- Older patients have a higher risk of death
- Sex does not affect survival in this study

Recommendations and Conclusion

Recommendations

- Encourage early HIV testing and start treatment early
- Give more attention to older patients because their risk is higher
- Focus prevention programs on major transmission groups
- Improve long-term follow-up and care for AIDS patients

Conclusion

- Survival probability decreases over time after diagnosis
- Many patients survive only a short period, while fewer survive many years
- Age is an important factor affecting survival
- Sex does not show a significant difference in survival
- Some transmission categories have different levels of risk

References

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Thank You

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